# Fayaz Ahmad Mir, PhD.

Nehs Neuroenergetics and Sleep Lab.
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**Professional Summary:**

- Ph.D. in Neurobiology and currently working as a postdoctoral fellow at Massachusetts General Hospital and Harvard Medical School, specializing in sleep and circadian neurobiology, mitochondrial metabolism, anesthesia-induced POCD, neurodegeneration, chemosensory neural networks and brain bioenergetic dynamics.
- Skilled in stereotaxic brain surgery in rodents, in vivo fiber photometry tracking spatiotemporal neuronal calcium and ATP dynamics with genetically encoded sensors, EEG/EMG electrode implantation, polysomnography, chemogenetics, and fluorescent/confocal microscopy, immunohistochemistry, and complex rodent behavioral paradigms.
- Analyzes EEG and fiber photometry data in MATLAB and Python, including automated sleep-wake scoring with AccuSleep in MATLAB and EEG spectral calculations in MATLAB. Presently building up data analysis pipelines for 2-photon imaging using EZcalcium (MATLAB) and Suite2p (Python).
- One of the lead postdocs on an NIH R01 project that investigated dietary intervention for aging and AD brain pathologies and anesthesia effects, author of 12 peer-reviewed publications and 3 book chapters, including 6 as first author.
- Elected to the membership of "Member Committee" Sleep Research Society, IL, USA
- Proficient in scientific and grant writing, with a consistent record of mentoring junior researchers in electrophysiology and in vivo imaging techniques.

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**Education Qualification:**

- **Ph.D. in Neuroscience** (2015 to 2022), School of Life Sciences, Jawaharlal Nehru University, New Delhi, India.
- **M.Phil., Neurobiology / Life Sciences** (2012 to 2014), Jawaharlal Nehru University, New Delhi, India.
- **Master of Science (M.Sc.), Biotechnology** (2010), Bangalore University, Karnataka, India.
- **Bachelor of Science (B.Sc.), Biotechnology** (2007), Bangalore University, Karnataka, India.

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**Postdoctoral Research Experience:**

**Massachusetts General Hospital, Harvard Medical School Boston, MA**

***Postdoctoral Research Fellow*** **01/2023-Present**

- Working on an active NIH R01 grant (PI: Dr. Christa J. Nehs, Mass General Brigham Department of Anesthesiology), directing aims on ketogenic and metabolic interventions to rescue sleep, memory and counter anesthesia-induced cognitive deficits and age-related neurodegeneration in rodent models.
- Built a multimodal in vivo pipeline (fiber photometry, wireless EEG/EMG, stereotaxic surgery, viral vector-mediated GECI expression, and molecular endpoints) to track brain ATP and calcium dynamics with GRAB-ATP1.0 and GCaMP6 sensors across sleep-wake states in aged rats and in AKAP11 (bipolar depression) and 3xTg Alzheimer's mouse models, linking brain energetics and circuit activity to mood, memory, and cognitive outcomes.
- Contributed to two completed manuscripts demonstrating (1) how a ketogenic diet improves sleep in aged female rats by lowering hippocampal oxidative stress and boosting cognition (Frontiers in Neuroscience, 2026, co-first author), and (2) another showing how KD and exogenous ketone supplementation blunts sevoflurane-induced burst

suppression, improves post-anesthesia cognition, and reduces cell-type specific oxidative damage in prefrontal cortex, a neuroprotective mechanism with direct perioperative relevance (under revision, Brain Research Bulletin, first author).

- Currently utilizing aged rats for assessing temporal ATP/calcium dynamics from PFC under baseline and after anesthesia or sleep deprivation conditions. Further, I am currently training in 2-photon microscopy and analysis skills, and I am already confident in performing data analysis using MATLAB and Python based libraries.
- Submitted (not funded) an NIH K99/R00 application built on a sleep-brain bioenergetic and chemosensory hypothesis of neuroprotection in aging and Alzheimer's models.
- Mentored around 10 summer fellows and research assistants in stereotaxic surgery, viral vector injections, EEG/EMG implantation in mice/rats, fiber photometry, and MATLAB-based signal processing.
- Presented my work at SfN 2025, two Gordon Research Conferences (Sleep Regulation and Function; Consciousness, Anesthesia and Evolutionary Biology), and an oral talk at the International Society of Sleep Research Annual Meeting.

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**School of Life Sciences, Jawaharlal Nehru University New Delhi, India**

**Graduate Research (PhD)** **2015 to 2022**

***Thesis: REM Sleep Regulation by Brainstem Chemosensory Networks: Role of Acid-Sensing Ion Channels and Proton Pumps in the Locus Coeruleus.*** *Advisor: Dr. Sushil K. Jha.*

- Designed and executed a full research project on REM sleep circuitry, from hypothesis generation through publication.
- Conducted stereotaxic surgeries, EEG/EMG polysomnography, immunohistochemistry, behavioral assays including contextual and cued fear conditioning, and pharmacological interventions in rats.
- First researcher to demonstrate that ASIC-2/3 channels in the locus coeruleus mediate hypercapnic modulation of NREM-to-REM sleep transitions (**Neuroscience Bulletin, 2021**).
- Discovered that the FDA-approved proton pump inhibitor lansoprazole suppresses locus coeruleus neuronal activity and increases REM sleep, a previously unrecognized sleep-promoting mechanism (**ACS Chemical Neuroscience, 2021**).
- Characterized the integrative role of KIR channels in the nucleus tractus solitarius in chemosensory-REM sleep homeostasis (**Scientific Reports, 2024**).
- Mentored $5^{+}$ master's and doctoral students in vivo electrophysiology, stereotaxic surgery, IHC, tissue histology and sleep scoring.
- Recipient, Senior Research Fellowship, Jawaharlal Nehru University.

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**Master's degree dissertation:**

**School of Life Sciences, Jawaharlal Nehru University** New Delhi, India

*M.Phil. Thesis: Role of Locus Coeruleus Proton Pumps and Na+/H+ Exchangers in REM Sleep Modulation in the Rat.*

- Investigated how proton pumps and $Na^{+}/H^{+}$ exchangers in the locus coeruleus regulate REM sleep, laying the groundwork for the doctoral thesis.
- Completed advanced coursework and laboratory rotations in neurobiology and life sciences, JNU, India.
- Prior research experience: Junior Research Fellow, CSIR-Institute of Himalayan Bioresource Technology, Palampur, India (04/2011 to 09/2011), conducting molecular biology work in plant biotechnology including DNA extraction, PCR, and gene expression analysis.
- Awarded the GATE Fellowship in Biotechnology (2012) and the CSIR Junior Research Fellowship (2011) based on competitive national examinations.

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**Grants & Funding:**

- **NIH R01 (Active)**: One of the lead Postdoctoral researchers working on funded research aims examining sleep, ketogenic metabolism, and anesthesia-induced neurodegeneration. PI: Dr. Christa J. Nehs, Massachusetts General Hospital / Harvard Medical School.
- **NIH K99/R00 (submitted but not funded)**: Central hypothesis: the ATP-sleep-chemosensory triad as a neuroprotective mechanism in aging, integrating fiber photometry, EEG spectral dynamics, and ketogenic interventions.
- Actively contributed to the NIH grant design, writing, yearly progress reports and significantly contributed to the innovative grant ideas presented in PIs NIH grants.
- Contributed to conceptualization, designing, writing and publication of the two first-author review publications in high-impact journals (ARR and Biobehavioral Neuroscience).

**Honors & Awards:**

- Sleep Research Society (SRS) Diversity Membership Award (2024 to Present)
- International Brain Research Organization (IBRO) Alumnus Award (2017)
- Senior Research Fellowship, Jawaharlal Nehru University (2013)
- GATE Fellowship in Biotechnology, Government of India (2012)
- CSIR Junior Research Fellowship, Council of Scientific and Industrial Research, India (2011).

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**Research Publications:**

1- **Mir FA**, et al. Ketogenic Diet Differentially Modulates Sevoflurane-Induced EEG Dynamics and Memory in Aged Male and Female Rats. Brain Research Bulletin (Under Review, Manuscript Number: BRB-D-26-01152).

2- Lark ARS†, **Mir FA**†, et al. Ketogenic interventions enhance REM sleep in females and support memory in aged rats. Front Aging Neurosci (**2026**).

3- Kang J, …, **Mir FA,** …, Nehs CJ. Gestational diabetes mellitus and adverse maternal and neonatal health outcomes: an umbrella review of meta-analyses. Obes Rev (**2026**); e70102.

4- Kang J, Kim HJ, Lark ARS, **Mir FA**, et al., Nehs CJ. Association of everyday discrimination with drug use during the COVID-19 pandemic in the All of Us Research Program. Psychiatry Investig (**2026**); 23(1):118-129.

5- Siegmann MJ, Zachs DP, Kenny JD, **Mir FA**, et al. Neurophysiological differences between dexmedetomidine sedation and natural sleep across the rodent lifespan: implications for aging and perioperative brain health. Anesthesiol Perioper Sci (**2026**); 4:10.

6- **Mir FA**, Jha SK, Nehs CJ. Noradrenergic dysregulation in the locus coeruleus: implications for neuropsychiatric disease pathophysiology, a systematic review. Neurosci Biobehav Rev (**2025**); 106442.

7- Kang J, …, **Mir FA**, …, Nehs CJ. Biomarkers for Alzheimer's disease across diverse biological domains: an umbrella review and evidence map. J Adv Res (**2025**).

8- Siegmann MJ, …, **Mir FA**, …, Nehs CJ. A ketogenic diet decreases sevoflurane-induced burst suppression in rats. Brain Res Bull (**2025**); 223:111274.

9- **Mir FA**, Lark ARS, Nehs CJ. Unraveling the interplay between sleep, redox metabolism, and aging: implications for brain health and longevity. Front Aging (**2025**); 6:1605070.

10- **Mir FA**, Jha SK. The Kir channel in the nucleus tractus solitarius integrates the chemosensory system with REM sleep executive machinery for homeostatic balance. Sci Rep (**2024**); 14:21651.

11- **Mir FA**, Amanullah A, et al. Neuroepigenetics of ageing and neurodegeneration-associated dementia: an updated review. Ageing Res Rev (**2023**); 91:102067.

12- **Mir FA**, Jha SK. Proton pump inhibitor lansoprazole inhibits locus coeruleus neuronal activity and increases REM sleep. ACS Chem Neurosci (**2021**); 12(22):4265-4274.

13- Mir FA, Jha SK. Locus coeruleus acid-sensing ion channels modulate sleep-wakefulness and state transition from NREM to REM sleep. Neurosci Bull (2021); 37(5):684-700.

14- Kant D, …, **Mir FA**, Jha SK. Effect of glial glutamine synthetase inhibition on recognition and temporal memories in the rat. Neurosci Lett (2014); 560:98-102.

**Book chapters**

1. Mir, F.A., et al. Chapter in Alzheimer's Disease Pathogenesis. Springer (2019).
2. Mir, F.A., et al. Chapter in Sleep, Memory & Synaptic Plasticity. Springer (2019).
3. Mir, F.A. Chapter in Encyclopedia of Animal Cognition and Behavior. Springer, Cham (2022).

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**Technical Skills:**

**In Vivo Electrophysiology:** Rodent stereotaxic surgery and microinjections, wireless and tethered EEG/EMG polysomnography, LFP acquisition, and sleep-wake scoring across NREM, REM, and wake states. Experienced with behavioral paradigms including novel object recognition, novel-arm Y-Maze, open field testing, and fear conditioning.

**Fiber Photometry & Advanced Imaging** In Vivo fiber photometry using GRAB-ATP1.0, iATPSnFR2 and GCaMP6 sensors to record real-time ATP and calcium dynamics in deep brain regions, applied across models of Alzheimer's disease, aging, and the AKAP11 mouse model of bipolar depression. Trained IHC, confocal microscopy, and widefield imaging for circuit-level and cellular characterization.

**Neuropharmacology & Circuit Manipulation:** Precision based drug infusions in deep brain regions like LC, NTS and RTN for modulation of neural circuits, stereotaxic based AAV viral vector delivery, and bilateral stereotaxic cannulation, paired with behavioral outcome assessment to connect circuit activity to cognitive and sleep phenotypes.

**Molecular & Cellular Biology** Immunohistochemistry, confocal imaging, Western blotting, qRT-PCR, primary cell isolation, and mammalian cell culture.

**Metabolic Phenotyping** Ketogenic diet and exogenous ketone oral gavage and tail vein blood glucose/ketone assessments to probe brain energetics and its role in cognitive resilience.

**Data Analysis & Software** MATLAB-based spectral and EEG signal-processing, sleep scoring and delta-power analysis pipelines, Python, GraphPad Prism, SigmaStat, and PRISMA-based systematic review methodology.Fiber photometry and 2 photon imaging data analysis in MATLAB/Python based functions and tools.

**Adobe Photoshop and BioRender:** Expert in graph/schematic model figure organization and editing for publication-ready images with adjusted text/legend/resolution.

**Other related tasks and Reviewer contributions:**

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- I have been invited to review many international journal publications and so I have reviewed 25+ manuscripts from MDPI and Frontiers journals.
- Invited reviewer, Oxford University Press, book proposal review for the Early Career Researcher First Book Prize (title: Neuroepigenetics: How We Live Shapes How We Age, 2026).
- Ad-hoc peer reviewer, Journal of Sleep and Vigilance, Springer (2023 to Present).
- Membership Committee Member, Sleep Research Society (May 2025 to Present); General Member since August 2024.
- Alumnus member, International Brain Research Organization (IBRO) network.

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**International conferences attended:**

- Society for Neuroscience (SfN) Annual Meeting, San Diego, CA, 2025. Poster: "Ketones modulate sleep and cognition in the aged brain."
- Gordon Research Conference on Sleep Regulation and Function, Lucca, Italy, 2025. Poster: "Dietary intervention improves sleep and cognition in old rats."
- Gordon Research Conference on Consciousness, Anesthesia and Evolutionary Biology, Galveston, TX and Easton, MA, 2024 to 2025. Poster presentation.
- International Society of Sleep Research (ISSR) Annual Meeting, India. Oral presentation.
- Harvard Medical School symposia (multiple, 2023 to 2025), building collaborative ties across MGH, Brigham and Women's Hospital, and Harvard-affiliated sleep research programs.

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**Workshops and training programs**

1. International Brain Research Organization (IBRO) training school and alumnus fellowship (2017).
2. Scientific Education and Research Board (SERB), 7$^{Th}$ Neuroscience School, organized by Department of Science and Technology, Govt. of India, December 16-30, 2013, at Central University of Hyderabad, Hyderabad-500 046 (India).
3. 8th International Conference of Laboratory Animal Scientists' Association (LASA), India on "Recent Advances in 3R's and Laboratory Animal Science", 25-26th November 2017, JNU, New Delhi, India.

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**References:**

1) Dr. Christa J. Nehs (**Postdoctoral Mentor**) Assistant Professor, Mass General Brigham Department of Anesthesiology, Massachusetts General Hospital, Harvard Medical School, Boston, MA.
✉CNEHS@mgh.harvard.edu ☎ +1-301-537-6902

2) Dr. Sushil K. Jha (**PhD Thesis Supervisor**) Associate Professor, School of Life Sciences, Jawaharlal Nehru University, New Delhi, India. ✉: sushilkjha@mail.jnu.ac.in ; sushil_1000@yahoo.com ☎ +91-9810376510

3) Dr. Eric Melonakos, Assistant Professor, Department of Cell Biology and Physiology, Brigham Young University.
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4) Dr. Arianna RS Lark, Senior Postdoctoral Fellow, Mass General Brigham Department of Anesthesiology, Massachusetts General Hospital, Harvard Medical School, Boston, MA.
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5) Dr. Mudasir Ahmad Khanday, Beth Israel Deaconess Medical Center, Harvard Medical School, Boston, MA
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6) Dr. Akash Gautam, Assistant Professor, Center for Neural & Cognitive Sciences, School of Medical Science University of Hyderabad, Hyderabad-500 046 (India).
✉akash@uohyd.ac.in ☎ +91-8179509014

7) Prof. B.N. Mallick, Professor of Neuroscience, Jawaharlal Nehru University, New Delhi, India.
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